



UnionPay Integrated Circuit Card Specifications

— Basic Specifications

Terms and Definitions

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Table of Contents

1	Application Scope.....	1
2	Normative references	2
3	Terms and definitions.....	3
	3.1 Terms concerning application layer.....	3
	3.2 Terms concerning communication layer.....	6
	3.3 Terms concerning security layer.....	8
4	Abbreviations.....	12

1 Application Scope

This book applies to all UPI Participants.

2 Normative references

The clauses in the following documents are incorporated by reference into this book. For dated references, all their subsequent modifications (excluding corrections) or revisions do not apply to this book. However, parties having reached an agreement based on this book are encouraged to study the latest versions of the referenced documents. For undated references, the latest versions apply to this book.

UnionPay Integrated Circuit Card Specifications – Basic Specifications (all parts)

3 Terms and definitions

The following terms and definitions are applicable to *UnionPay Integrated Circuit Card Specifications*.

3.1 Terms concerning application layer

Application

Application protocols and relevant data sets which are used between cards and terminals.

Application file locator

Locator that indicates the location and record range of application –related files.

Application interchange profile

Indicator of functions supported by cards.

Byte

Consists of 8 bits and designated from b1 to b8; from the least significant bit (LSB, b1) to the most significant bit (MSB, b8).

Card

Cards issued by the bank that support payment functions.

Clearing

A process where the Issuer processes the transaction data submitted by the Acquirer. To the Issuer, clearing provides another chance to synchronize the back-office counter with the counter embedded in cards. To the Acquirer, clearing is to submit the transaction data to the Issuer.

Command

A message sent from the terminal to IC cards to initiate one operation or request a response.

Consumer device

Cards used by consumers to make transaction payments (PICC) or other devices containing transaction chips (such as mobile phones or PDA).

Consumer device cardholder verification method (CDCVM)

A cardholder verification method based on mobile payment devices, which is only enabled in qUICS process.

Electronic cash (EC)

Function supporting low-value payment based on debit/ credit application.

Financial transaction

Information exchanges and funds arising from clearing/settlement among the Cardholder, Issuer, Merchant and Acquirer that result in commodity or service exchanges between the Cardholder and the Merchant.

Function

A sequence combined by one or more commands, which complete the entire (or part of the) transaction.

Integrated Circuit (IC)

Electronic components boast processing and/or storage functions.

Integrated Circuit (s) Card (ICC)

A card into which one or more integrated circuits are inserted to perform processing and memory functions

Interface device

The area where an IC card is inserted in a terminal.

Issuer action code

An issuer-determined action code that is used to determine actions in conjunction with TVR.

Load

A process that increases the electronic cash balance in the card. The loading may be carried out in multiple ways, such as transferring funds from the master account or depositing cash or transferring funds from other accounts. However, the electronic cash balance should not exceed the electronic cash balance limit after loading.

Magstripe

Magnetic Stripe that contains coded information.

Message

Byte strings that do not contain transmission control characters sent from terminals to cards, or vice versa.

Message authentication code

Codes mainly used to verify the integrity of message, which is generated by computation of transaction data and relevant parameters.

Offline data authentication for online authorization (ODA for online authorization)

A feature proposed for quick entry and exit in mass transit network, which allows a contactless terminal to quickly authentic the card offline and sent the transaction data for deferred authorization.

Payment account reference (PAR)

A card data element (tag '9F24') used to associate different Tokens deriving from the same bankcard. The same PAN and its Tokens shall have one identical PAR.

Path

An application path that is selected based on the terminal-supported magnetic stripe data format, or quick debit/credit application, which determines the action of the card.

Payment system environment

A set of logical conditions established within IC cards after payment system application in line with *UICS* has been chosen.

Response

A message returned to the terminal after the IC card has completed the task required by the command message.

Script

Command or sequence of commands sent from the Issuer to terminals, with the purpose of consecutively inputting commands to IC cards.

Terminal

Devices installed at transaction venues to complete financial transactions with the support of IC cards. It includes interface devices and other components and interfaces (such as those supporting communication with the host).

Terminal action code

Acquirer-determined action code that is used to determine actions in conjunction with TVR,

Transaction declined

The regulated procedures have been completed, however, the card or terminal or the Issuer does not allow the transaction to occur due to some reasons (including without limitation, risk management, business rules and policy considerations).

Transaction log

Detailed records of recent transactions, from which, the history of cards' transactions can be learned.

Transaction terminated

The regulated transaction procedures failed to be completed and the transaction could not be continued due to some reasons (including without limitation, lack of necessary data or failure of the terminal or card).

3.2 Terms concerning communication layer

Anti-collision loop

An algorithm PCD used to communicate with one or more PICCs in its excitation field.

Binary phase shift keying (BPSK)

A 180° phase shift keying, resulting in two possible phase statuses.

Bit collision detection protocol

An anti-collision method detects bit-level collisions within a frame. A collision occurs when at least two PICCs send complementary bit patterns to PCD. In this case, bit patterns are merged. During the entire (100%) bit duration, sub-carrier is used for modulation.

Bit duration

Active time of a logical status. When the time ends, a new bit will start.

Block

A special type of frame indicates the valid protocol data format.

Note: a valid protocol data format includes I-block, R-block or S-block.

Byte

Consisting of 8 bits and designated from b1 to b8, from most significant bit (MSB, b8) to the least significant bit (LSB, b1).

Collision

Within the same time period and the working field of one PCD, two or more PICCs are being transmitted. In this case, the PCD is unable to tell which PICC data comes from.

Contactless integrated circuit(s) card

A type of ID-1 card (as regulated in ISO/IEC FDIS 7810) with integrated circuits sealed inside. The communication with the integrated circuits is done through contactless.

Elementary time unit (etu)

In this section, the elementary time unit (etu) is defined as: $1\text{etu}=128/f_c$.

Epilogue field

The last part of the block that includes the error detection code (EDC).

Frame

A series of data bits and optional error detection bits, accompanied with start and end delimiters. (Note: Type A PICCs use standard frames defined for Type A PICCs; Type B PICCs use standard frames defined for Type B PICCs.)

Higher layer

Part of an application or higher layer protocol that is not described in this section.

Invalid block

A frame that includes an invalid protocol format.

Note: The frame cannot be interpreted as an invalid block without receiving the frame timeout.

Modulation index

Defined as $[a-b]/[a+b]$, of which, a is the peak value of signal amplitude and b is the minimum value of signal amplitude.

Non-return to zero (NRZ-L)

A coding scheme using one of two physical statuses defined by communication media to express the logic status during bit duration.

Prologue field

The first part of the block that includes protocol control bytes (PCB) (not using CID and NAD defined in UICS.)

Proximity Coupling Device (PCD)

Using inductive coupling to provide energy to PICCs, and control the reading/writing device which exchanges data with PICCs.

Proximity IC card (PICC)

A type of ID-1 card, with integrated circuits and coupled circuits sealed inside, where the communication with integrated circuits is done through inductive coupling of the proximity-coupling device.

Reader

The reader often comes in two types of contactless transactions:

- As a detached reader that has a communication link with the POS.
- As a reader incorporated into the POS equipment.

The first type is often adopted in MSD method, which is however not regulated in this section.

Unless otherwise specified, the “Reader” in this section includes two types, without specifically specifying in which physical module (reader or POS equipment) an operation is performed.

Subcarrier

RF signals generated when modulating carrier frequency f_c based on frequency f_s .

Time slot protocol

A method used by PCD to establish logical channels with one or more PICCs. This method, like the slotted-Aloha method, uses time slot for positioning a PICC response.

Unique identifier (UID)

A serial number required by anti-collision algorithm of Type A.

3.3 Terms concerning security layer**Accelerated revocation**

Revoke a key prior to the announced expiry date.

Asymmetric cryptographic technique

A cryptographic technique uses two related transformations: a public transformation (defined by the public key) and a private transformation (defined by the private key). The two transformations have the property that, given the public transformation, it is computationally infeasible to derive the private transformation through computation.

Authentication

A measure to certify the identity declared of an entity.

Cardholder verification method

A method used at terminals to verify that the cardholder is legal and the card is not lost or stolen.

Certificate

Unforgeable data that are generated when the authentication authority issuing the certificate uses its private key to sign the public key, identity and other relevant information of an entity.

Certification authority

A credible third party that certifies the ownership of the public key and other relevant information.

Chinese remainder theorem (CRT)

A special representation of RSA private key that may boost signature computation speed.

Compromise

A settlement where two sides make concessions.

Concatenation

Two elements are connected by adding the bytes of the second element to the end of the first element. The order of the bytes of each element in the resultant string

remains the same as that sent from the IC cards to terminals, namely, the high bytes first. Within each byte, the bits are arranged in descending order.

Cryptographic algorithm

A transformation algorithm to conceal or display the information contents.

Cryptogram

Results of encrypted computation.

Cryptoperiod

A timeframe when a key is active, or a period when a key is valid in a given system.

Decipherment

The inverse operation of an encryption process.

Digital signature

An asymmetric encryption transformation of data that allows the recipient to confirm the source and integrity of data, and protects the data sent by the sender and received by the recipient from being falsified by a third party. It also protects the data sent by the sender from being falsified by the recipient.

Encipherment

A reversible transformation of data based on a cryptographic algorithm so as to generate cryptogram.

Hash function

A function mapping the bit strings of variable length to bit strings of a fixed length that meets the following two conditions:

- For a given output, it is infeasible to derive the corresponding input data;
- For a given input, it is infeasible to derive another input with the same output through computation.

In addition, if a hash function is required to have anti-collision functions, the following conditions will be met:

- It is infeasible to find two different inputs with the same output.

Hash result

Output bit strings of a hash function.

Key

A symbol sequence controlling the encryption transformation operations.

Key expiry date

A deadline when the signature generated by a key will be no longer valid. The issuer certificate signed by this key must expire on or before this day. Following this day, the key may be terminated from terminals.

Key introduction

The process generates, distributes, and uses a key pair.

Key life cycle

All phases of key management, including key planning, generation, revocation, destroy, and filing.

Key replacement

To revoke a key and at the same time, introduce a new key to replace it.

Key revocation

A key management process of revoking the key in use and handle the leftover problems after its use. The key may be revoked according to the plan or revoked ahead of schedule.

Key revocation date

Following this day, any and all legal cards in use will not contain the certificate signed by the key. Therefore, the key may be deleted from terminals. As to a key revoked according to the plan, the key revocation date shall be same as the key expiry date.

Logical compromise

Compromise of a key due to improvement of cryptanalysis techniques and/or computing power.

Padding

Adding overhead bits to one end of a data string.

Physical compromise

Compromise of a key due to lack of security protection or theft of hardware security module or unauthorized access, etc.

PIN pad

A set of numeric and command keys to input personal identification codes.

Plaintext

Un-encrypted clear data.

Planned revocation

Revoke a key according to the announced key expiry date.

Potential compromise

Improved cryptanalysis techniques and/or computing power that can lead to the compromise of keys of a certain length.

Private key

A dedicated key belonging to an entity with asymmetric key pair. In the case of a digital signature scheme, the private key defines the signature function.

Public key

A key belonging to an entity with asymmetric key pair that can be made public. In the case of a digital signature scheme, the public key defines the verification function.

Public key certificate

The public key information of an entity signed by the certification authority and thereby rendered unforgeable.

Secret key

A secret key created with symmetrical encryption techniques and used by a designated entity only.

Symmetric cryptographic technique

The sender and the receiver use the same secret key in the encryption of the data transformation. Without the secret key, it is infeasible to derive the data transformation of both sides.

4 Abbreviations

$A \bmod n$	The remainder from a division of n by A , i.e. the sole integer r , $0 \leq r < n$, with another integer, $A = dn + r$.
$A := B$	A is assigned the value of B .
$A = B$	Value A equals Value B .
$A \equiv B \bmod n$	Integer A is coresidual to Integer B relative to $\bmod n$, i.e. there is an integer d that fits $(A - B) = dn$.
AAC	Application Authentication Cryptogram
AAR	Application Authorization Referral
AC	Application Cryptogram, applicable to Part 4: Specification for Application of Debit/Credit
AC	Anti-collision, applicable to Part 11: Specification for Communications with Contactless IC Card
ACK	Positive Acknowledgement
ADA	Application Default Action
ADC	Application Data Coding, Type B
ADF	Application Definition File
AEF	Application Elementary File
AFI	Application Family Identifier, Type B
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
AM	Amplitude Modulation
APDU	Application Protocol Data Unit
Apf	Anti-collision Prefix f, used in REQb/WUPb, Type B
Apn	Anti-collision Prefix n, used in Slot-MARKER Command, Type B
ARC	Authorization Response Code
ARPC	Authorization Response Cryptogram
ARQC	Authorization ReQuest Cryptogram
ASK	Amplitude Shift Keying
ATC	Application Transaction Counter
ATM	Automated Teller Machine
ATQ	Answer To reQuest

ATQA	Answer To reQuest, Type A
ATQB	Answer To reQuest, Type B
ATS	Answer To Select, Type A
ATTRIB	PICC selection command, Type B
AUC	Application Usage Control
AuthC	Authorization Controls
b	Binary
BCC	(Type A) UID CLn check byte, calculated as exclusive-or over the 4 previous bytes, Type A.
BCD	Binary Coded Decimal
BER	Basic Encoding Rules
BIN	Bank Identification Number
BPSK	Binary Phase Shift Keying
C: =(A B)	Link m-digital Number B with n-digital A, defined as: $C=2mA+B$
CA	Certificate Authority
CAM	Card Authentication Method
CBC	Cipher Block Chaining
CDA	Combined DDA/AC Generation
CDOL	Card Risk Management Data Object List
CID	Cryptogram Information Data
CLA	Class Byte of the Command Message
CLn	Cascade Level n, Type A, $3 \geq n \geq 1$
C-MAC	Command-Message Authentication Code
cn	Compressed Numeric
CRC	Cyclic Redundancy Check
CRC_A	Cyclic Redundancy Check error detection code A
CRC_B	Cyclic Redundancy Check error detection code for Type B
CT	Cascade Tag, Type A
CTTA	Cumulative Total Transaction Amount
CTTAL	Cumulative Total Transaction Amount Limit
CTTAUL	Cumulative Total Transaction Amount Upper Limit
CVM	Cardholder Verification Method

CVN	Card Verification Number
D	Divisor
dCVN	Dynamic Card Verification Number, a dynamic signature generated with the card application. The dynamic CVN replaces the static CVN of the equivalent data in disk track 2.
DDA	Dynamic Data Authentication, can prevent false card in offline authentication, the card generates RSA signature against specific data of the transaction for terminal verification.
DDF	Directory Definition File
DDOL	Dynamic Data Authentication Data Object List
DEK/TK	Data Encryption Key
DES	Data Encryption Standard
DF	Dedicated File
DGI	Data Grouping Identifier
DIR	Directory
DKI	Derivation Key Index
DOL	Data Object List
DR	Divisor Receive (PCD to PICC)
DRI	Divisor Receive Integer (PCD to PICC)
DS	Divisor Send (PICC to PCD)
DSI	Divisor Send Integer (PICC to PCD)
E	End of communication, Type A
EC	Electronic Cash
ECB	Electronic Code Book
EDC	Error Detection Code
EF	Elementary File
EGT	Extra Guard Time, Type B
EMV	Europay, MasterCard and VISA
ENC MDK	Master Data encipherment DEA Key
EOF	End Of Frame
EoS	End of Sequence
etu	Elementary time unit
fc	Carrier frequency

FCI	File Control Information, response and return with card when the reader or terminal selects card application. (use Select command)
fDDA	Fast DDA, a fast DDA that fits the definition of UnionPay Integrated Circuit Card Specifications, is used in qUICS trading, allows the reader to issue READ RECORD command to get access to DDA related data, and then implement DDA computing when the card is removed from the induction area.
FDT	Frame Delay Time
FIPS	Federal Information Processing Standard
FO	Frame Option, Type B
fs	Frequency of subcarrier modulation
FSC	Frame Size for proximity Card
FSCI	Frame Size for proximity Card Integer
FSD	Frame Size for proximity coupling Device
FSDI	Frame Size for proximity coupling Device Integer
FWI	(Frame Waiting Time)—An integer coding used to define the time period, which defines the longest response time of the PICC after the end of frame of the reader.
FWT	Frame Waiting Time
FWTTEMP	Temporary Frame Waiting Time
GPO	Get Processing Options
H:=Hash[MSG]	Conduct Hash algorithm on message in any length with 160-digital Hash function.
Hex	Hexadecimal
HLTA	Halt Command, Type A
HLTB	Halt Command, Type B
HSM	Hardware Secure Module
IAC	Issuer Action Code, used to determine the conditions of IAC Denial, IAC Online and IAC Default.
IC	Integrated Circuit
ICC	Integrated Circuit(s) Card
iCVN	Alternative CVN, used for mirroring of equivalent data in track 2 for personalization.
ID	Identification number

IDD	Issuer Defined Data
IEC	International Electrotechnical Commission
IFD	Interface Device
INF	Information field
INS	Instruction Byte of Command Message
ISO	International Organization for Standardization
ISS	ISSuer
KDEK	Sole Secret Key for the card, used to generate DES Secret Key or other Session Key for optional confidential data.
KEK/TK	Key Exchange Key Transport Key—shared by data preparation system and personal equipment.
KEKISS	Key Exchange Key—shared by issuer and data preparation system.
KENC	Sole Secret Key for the card, used to generate encrypted Session Key.
KMAC	Sole Secret Key for the card, used to generate C-MAC Session Key.
KMC	Master DES Key, used to distribute Secret Key to generate KENC, KDEK and KMAC in the process of personalization.
KMCID	Identifier of the Master Key for Personalization
KS	Session Key
Lc	Exact Length of Data Sent by the TAL in a Case 3 or 4 Command.
LDD	Length of the ICC Dynamic Data.
LRC	(Longitudinal Redundancy Check) Used to verify data to secure no data is lost when the data are read out of the magnetic stripe.
LSB	Least Significant Bit
LV	Low Value
M	Mandatory
MAC	Message Authentication Code
MAC MDK	Master DEA Key for message authentication code
MAC UDK	Unique DEA Key for message authentication code
max	Index to define a maximum value
MBL	Maximum Buffer Length
MBLI	Maximum Buffer Length Index
MDK	Master DEA Key, a double-length DES Key used for derivation of the unique key for card for online verification of card.

MF	Master File
min	Index to define a minimum value
MMYY	Month, Year
MSB	Most Significant Bit
MSI	Magnetic Stripe Image, not applicable to contactless magnetic stripe technology.
n	Numeric
N	Number of anticollision slots or PICC response probability in each slot, Type B.
N/A	Not Applicable
NAD	Node Address
NAK	Negative Acknowledgement
NCA	Length of the Certification Authority Public Key Modulus
NI	Length of the Issuer Public Key Modulus
NIC	Length of the ICC Public Key Modulus
NRZ-L	Non-Return to Zero, (L for level)
NVB	Number of Valid Bits, Type A
O	Optional
OOK	On/Off Keying
OSI	Open System Interconnection
P	Odd Parity Bit, Type A
P1	Parameter 1
P2	Parameter 2
PAN	Primary Account Number
PARAM	PARAMeter
PCA	Certification Authority Public Key
PCB	Protocol Control Byte
PCD	Proximity Coupling Device, a reader that uses inductive coupling to provide power to consumers' device and controls exchange of data with consumers' device.
PDA	Personal Digital Assistant
PDOL	Processing Options Data Object List, a list of terminal data object required for card.

PEK/TK	A dedicated transmission key for PIN transmission
PI	Issuer Public Key
PIC	ICC Public Key
PICC	Proximity IC Card, identification card operated on coupling equipment, type ID-1 card (larger size card)
PIN	Personal Identification Number
PIX	Proprietary Application Identifier Extension
PKI	Public Key Infrastructure
PM	Phase Modulation
PPS	Protocol and Parameter Selection
PPS0	Protocol and Parameter Selection parameter 0
PPS1	Protocol and Parameter Selection parameter 1
PPSE	Proximity Payment Systems Environment, a list of supported application identifier, application label and application priority indicator, can be accessed through contactless interface. The list encompasses entries of all directories, returned from FCI as responded by SELECT PPSE (“2PAY.SYS.DDF01”).
PPSS	Protocol and Parameter Selection Start
Proximity	In this document, proximity means contactless technology described in UICS.
PSE	Payment System Environment
PUPI	Pseudo-Unique PICC Identifier, Type B
PVKI	PIN Verification Key Index
qUICS	Quick UICS securing quick transaction through contactless interface.
R	Slot number chosen by the PICC during the anti-collision sequence, Type B.
R(ACK)	R-block containing a positive acknowledge
R(NAK)	R-block containing a negative acknowledge
RATS	Request for Answer To Select, Type A
REQA	ReQuest command, Type A
REQB	ReQuest Command, Type B
RF	Radio Frequency
RFU	Reserved for Future Use

RID	Registered Application Provider Identifier
RSA	An asymmetric cryptographic algorithm raised by Rivest, Sharmir and Adleman, used for encipherment and verification.
S	Start of communication, Type A
SAD	Signed Static Application Data
SAK	Select AcKnowledge, Type A
SCA	Certification Authority Private Key
SDA	Static Data Authentication
SDAD	Signed Dynamic Application Data
SEL	SElect code, Type A
SFGI	Start-up Frame Guard time Integer
SFI	Short File Identifier
SHA	Secure Hash Algorithm
SI	Issuer Private Key
SIC	ICC Private Key
SKUENC	Used for encrypted Session Key, generated by KENC
SKUKEK	Used for encrypted DES Key or other Session Key for confidential data, generated by KDEK.
SKUMAC	Used to create C-MAC in the processing of command, generated by KMAC.
SOF	Start Of Frame
SoS	Start of Sequence
Status Check	Status check adopted by some merchants (for example gas stations) for online verification of a specific amount, implemented with one currency.
SUDK ENC	Encrypted session key for proprietary data generated with MAC ENC.
SUDK MAC	MAC Session key generated with MAC UDK
SW1	Status Word One
SW2	Status Word Two
TAC	Terminal Action Code
TAL	Terminal Application Layer

TC	Transaction Certificate, encrypted document generated in the acceptance of transactions.
TDOL	Transaction Certificate Data Object List
TK	Transmission Key
TLV	Tag Length Value
TR0	Guard Time, Type B
TR1	Synchronization Time, Type B
TSI	Transaction Status Information
TVR	Terminal Verification Results
UDK	(Unique DEA Key), the sole double-length DES Key, used for online verification of card.
UID	Unique Identifier, Type A
uidn	Byte number n of UID, Type A
WTX	Waiting Time eXtension
WTXM	Waiting Time eXtension Multiplier
WUPA	Wake-UP command, Type A
WUPB	Wake-UP command, Type B
$X := \text{Recover}(\text{PK})[Y]$	Use Public Key to restore data block Y with asymmetric reversible algorithm.
$X := \text{ALG-1}(K)[Y]$	Use Secret Key K to decrypt 64-digital or 128-digital Data Block Y with the 64-digital or 128-digital block encryption method.
$Y := \text{ALG}(K)[X]$	Use Secret Key K to encrypt the 64-digital or 128-digital Data Block X with the 64-digital or 128-digital block encryption method.
$Y := \text{Sign}(\text{SK})[X]$	Use Private Key SK to sign the Data Block X with asymmetric reversible algorithm.